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Design of a linear-motion dual-stage actuation system for precision control

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Abstract
Actuators with high linear-motion speed, high positioning resolution and a long motion stroke are needed in many precision machining systems. In some current systems, voice coil motors (VCMs) are implemented for servo control. While the voice coil motors may provide the long motion stroke needed in many applications, the main obstacle that hinders the improvement of the machining accuracy and efficiency is their limited bandwidth. To fundamentally solve this issue, we propose to develop a dual-stage actuation system that consists of a voice coil motor that covers the coarse motion, and a piezoelectric stack actuator that induces the fine motion, thus enhancing the positioning accuracy. The focus of this present research is the mechatronics design and synthesis of the new actuation system. In particular, a flexure hinge based mechanism is developed to provide a motion guide and preload to the piezoelectric stack actuator that is serially connected to the voice coil motor. This mechanism is built upon parallel plane flexure hinges. A series of numerical and experimental studies are carried out to facilitate the system design and the model identification. The effectiveness of the proposed system is demonstrated through open-loop studies and preliminary closed-loop control practice. While the primary goal of this particular design is aimed at enhancing optical lens machining, the concept and approach outlined are generic and can be extended to a variety of applications.

(Some figures in this article are in colour only in the electronic version)

1. Introduction

Precision control systems with high motion speed, high positioning resolution and a long motion stroke are needed in industrial applications, e.g., various machining processes and computer components. However, when only a single actuator is employed, it is often difficult to simultaneously satisfy all the ever-increasing performance demands. For example, in the machining of optical lenses with irregular surfaces, a cutter tool feed drive system is employed to provide a linear motion with a long stroke (at the centimeter level) and high precision (at the sub-micrometer level or even higher) simultaneously. Currently, in these systems, linear motors such as voice coil motors are implemented for servo control. Although exhibiting improved features, including direct transmission and non-backlash, over previous servo motors, voice coil motors still have noticeable bandwidth limitations (<100 Hz) and positioning resolution limitations ($\sim 1\ \mu$m). These inherent limitations have hindered the further improvement of machining performance.

A dual-stage actuation concept has been developed in the computer hard disk drive (HDD) industry. In an HDD system, a voice coil motor is generally used to position the read/write head onto the spinning disk. To increase the storage density, the precision of such a servo system needs to be significantly increased. As a result, a series of studies have proposed adding a second stage actuator, e.g., a piezoelectric actuator, to increase the servo bandwidth and precision [1, 2]. Although a single piezoelectric actuator cannot alone provide an adequate motion stroke under the design limitations in a HDD, it can compensate for the error of the voice coil motor. Following mechatronics synthesis studies, several closed-loop control algorithms have been formulated for precision tracking control in HDDs using such a dual-stage actuation concept [3–5].

The dual-actuation idea has also been practiced in macro/microrobot applications such as space robots and medical robots [6–8], where a macrorobot provides the large workspace and a microrobot (fixed onto the macro one) provides the high precision adjustment of the position and posture. There have been recent efforts to apply the dual-stage concept in the precision positioning system [9, 10]. Liu *et al* proposed a high precision table/platform consisting of a voice coil motor and a piezoelectric actuator [9]. In their system, both ends of the piezoelectric actuator were fixed to a connection rod between the moving shaft of the voice coil motor and a hammer, and the hammer pushes the table to the target position. Their experimental results indicated that the system reached a positioning accuracy of 10 nm. Chen and Dwang [11] proposed a dual-actuation system, which comprised a servo motor-actuated ballscrew mechanism and a piezoelectric stack nut. In order to achieve a high precision, the piezoelectric stack was embedded in a nut with a preload to eliminate the backlash. This piezoelectric nut can actively adjust the ballscrew preload, and also yield precision control of the motion stage. Elfizy *et al* [12] developed a dual-stage feed drive system for milling, in which a linear motor and a piezoelectric stack were used respectively as the coarse and fine actuators. The dual-stage system achieved a 75% reduction in tracking error in comparison to the single linear motor controlled drive. They further extended the dual-stage idea to a planar configuration. In sinusoidal profile cutting experiments, the maximum tracking error was reduced by 83% and the average magnitude of the error was reduced by 63%. Similarly, Tong *et al* [13] utilized a linear motor and a magnetostrictive actuator for dual-stage control. A coarse/fine dual-stage manipulator for micro-tele-operation was reported [14], in which an XY fine stage was mounted on a 2D coarse positioner. The coarse and fine positioners were complementary to each other, so the integral system can realize a high precision operation in a large motion workspace. Kim *et al* [15] reported their dual-actuation research application on cam machining, in which an electro-hydraulic actuator with a 25 mm gross motion and a piezoelectric actuator with a 40 μm fine motion are integrated together. In their experiment of tracking a cam profile at a rate of 600 rpm, the proposed dual-stage system generated a tracking error of 4 μm peak-to-valley with an 0.80 μm RMS value, showing a substantial improvement over the 16 μm peak-to-valley and 2.64 μm RMS errors generated by the electro-hydraulic servo system alone. Kim *et al* [16] developed a dual-actuation system for a flat panel display process, in which the coarse stage is actuated by a linear motor and the fine stage is actuated by a voice coil motor. Due to the advantages of the dual-stage actuation, the integral system has a 1.2 m motion stoke and 10 nm resolution. Park [17] proposed a three degree-of-freedom precision hybrid stage, which is driven respectively by three linear motors (macro-actuators) and four voice coil motors (micro-actuators) to realize coarse and fine positioning. Their preliminary results showed that the system has a motion range of 400 mm with nanometer level resolution (10 nm) and repeatability (50 nm).

The primary goal of this research is to develop a prototype for a dual-stage actuation system that can be used for cutter tool control with a linear, long motion stroke and high precision. Our basic idea follows the previous research on the dual-stage actuation concept, i.e., using a voice coil motor for the long stroke motion while employing a piezoelectric stack for high precision compensation. The two actuators are connected serially. In order to take full advantage of the dual-stage concept, a flexure fixture is designed as the connection mechanism that can lead to an adequate open-loop bandwidth, and at the same time maintain the motion stroke of the piezoelectric stack actuator. This fixture, which houses the piezoelectric actuator, features a parallel flexure hinge structure that can provide the preload and motion guide for the piezoelectric stack actuator. Correlated finite element analysis and experimental studies are carried out for system design and synthesis. The performance of the dual-stage actuation system is demonstrated experimentally by open-loop control and preliminary closed-loop control investigations. System identification using frequency domain information is carried out, and the close-loop control is built upon a dual-loop strategy based on PID, where the piezoelectric loop is used to compensate the voice coil motor error. While the work performed in this research is targeted at the specific requirements for a linear-motion control system to be used in optical lens machining, the design methodology can be generalized to generic dual-actuation systems for a variety of applications.

2. Dual-actuation system design

2.1. Design objective

In many machining/manufacturing processes, the part to be machined (e.g., a lens in optical machining) is mounted on a spindle and subject to a constant-speed rotation, and a linear-motion control system positions the cutter tool with a pre-specified trajectory to facilitate the surface cutting. The goal of this research is to develop a dual-actuation system that can simultaneously provide a long motion stroke (at the centimeter level) and high positioning accuracy (at the sub-micrometer to nanometer level) for cutter motion control. This will be accomplished by integrating a piezoelectric stack actuator (fine actuator) with a voice coil motor (VCM, coarse actuator). In other words, the motion range of the piezoelectric actuator will cover at least the motion error of the VCM, and the piezoelectric actuator's bandwidth is high enough to compensate the system error. Since high throughput is generally needed in manufacturing, the control system should be effective at high machining speed. Meanwhile, the system should provide a high stiffness (especially in the motion direction) and a continuous stall force, which can reduce the disturbance effect during the cutting process. To help reduce the effect of an external disturbance under operating conditions, the minimum natural frequency of the structural components should be high (>1000 Hz in this research) to avoid low-frequency resonances.

2.2. Design overview

Figure 1 presents a solid model view of the configuration of the dual-stage actuation system. A linear VCM is used as the coarse actuator. The stator of the VCM is fixed on the base, while the output coil of the motor is supported at the

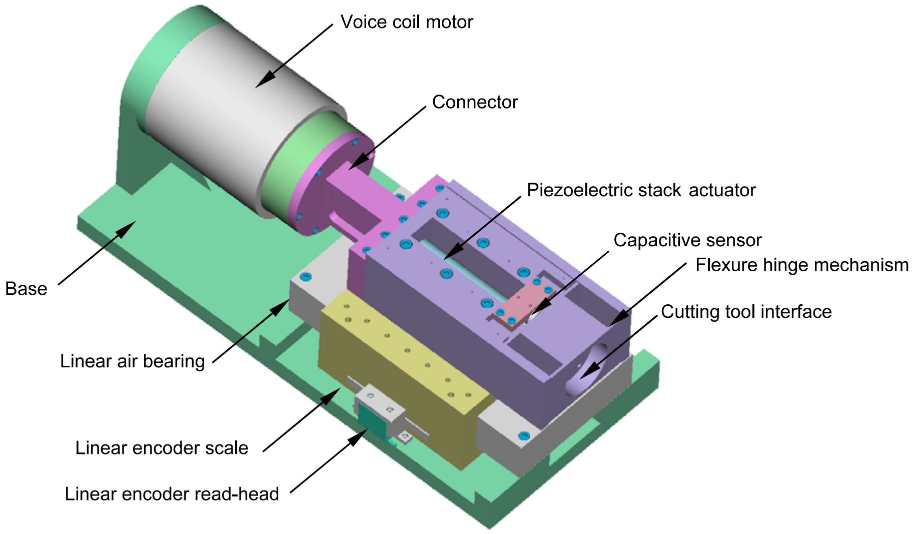

Figure 1. Design overview of the dual-stage actuation system.

other end by a linear air bearing via a connector between the coil and the sliding bar of the air bearing. A piezoelectric stack actuator is housed inside the flexure mechanism under preload. The flexure mechanism is also fixed on the sliding bar of the air bearing. The cutting tool will be installed on this fixture as well. In this system, the coarse VCM actuator pushes the sliding bar of the air bearing, and the fine piezoelectric actuator induces a motion relative to this sliding bar for compensation. As suggested by previous studies [18], a high precision air bearing can improve the system, supporting stiffness and guarantee a straight linear guide for the VCM output. It is worth mentioning that one key component in this design is the flexure mechanism that houses the piezoelectric stack actuator. In this research, in order to provide a preload and motion guide for the linear-motion output, four parallel beam flexure hinges are used to form a monolithic structure that yields a seamless connection between the piezoelectric stack and the motion output tip, which will be further discussed in section 2.3.

To realize the real time control of the dual-actuation system, two sensors are employed in this design. A linear encoder sensor is used to measure the motion of the air bearing's sliding bar, which reflects the motion of the voice coil motor. For the fine stage piezoelectric actuator, a capacitive sensor is mounted inside the flexure fixture to measure the micromotion of the piezoelectric stack actuator.

2.3. Flexure mechanism analysis

In this research, we use a high-voltage piezoelectric stack actuator (A050120, Kinetic Ceramics Inc) as the fine stage actuator. This stack actuator has a 120 μm motion stroke and a 4500 N output force. The nominal bandwidth of this stack actuator is 12.3 kHz. The actual bandwidth that can be realized in the system depends on the motion magnitude, due to the amplifier factor. In order to take full advantage of the motion stroke of the piezoelectric stack actuator, as well as its actuation bandwidth, the connection fixture (between the actuator and the cutting tool) should be well aligned, lead to a minimized loss of displacement stroke, and maintain the necessary rigidity in all directions to reduce vibration within the operating frequency range. We develop a flexure hinge mechanism to accomplish the design objective.

Flexure hinge based mechanisms are frequently used from motion guiding in various applications [19, 20]. Here we adopt a simple beam-type flexure hinge (figure 2(a)). This type of flexure hinge is easy to fabricate, and can provide the one degree-of-freedom motion guide necessary to the system. In particular, we propose parallel beam-type flexure hinges (figure 2(b)) in this research, since this design can minimize the deformation in the non-motion directions induced by cutting forces during the machining process. Generally, the cutter directly generates three-dimensional cutting forces upon the cutting tool tip (figure 2(c)). The cutting is facilitated by the Y-axis motion of the cutter tool, which is subject to linear-motion control by the dual-stage actuation system. The X-axis and Z-axis, however, are not subject to active control. Hence, high stiffness is required for these two directions to resist the cutting force disturbance. Meanwhile, the cutting force acting on the cutter tip also generates additional deformations of Y–Z twisting and X–Y torsion, since the cutter is a cantilever structure with respect to the flexure hinge mechanism. These unwanted deformations could degrade the final machining precision.

Based on the above observations, we adopt a layout with a parallel beam-type flexure hinge. This design can increase the linear stiffness as well as the torsional stiffness in the aforementioned non-motion directions. We use ANSYS to perform numerical analysis and design. The flexure hinge mechanism is made of aluminum alloy 2024 T3 with mass density 2700 kg m^{-3}, Poison's ratio 0.30 and Young's modulus 7.4×10^{10} N m^{-2}. The analysis results reported hereafter are based on the flexure hinge profile 25 mm $\times$ 2.5 mm $\times$ 66 mm, which is the actual design dimension that is reached after design iteration. We first analyze the static deformation of the entire flexure hinge mechanism/housing under the action

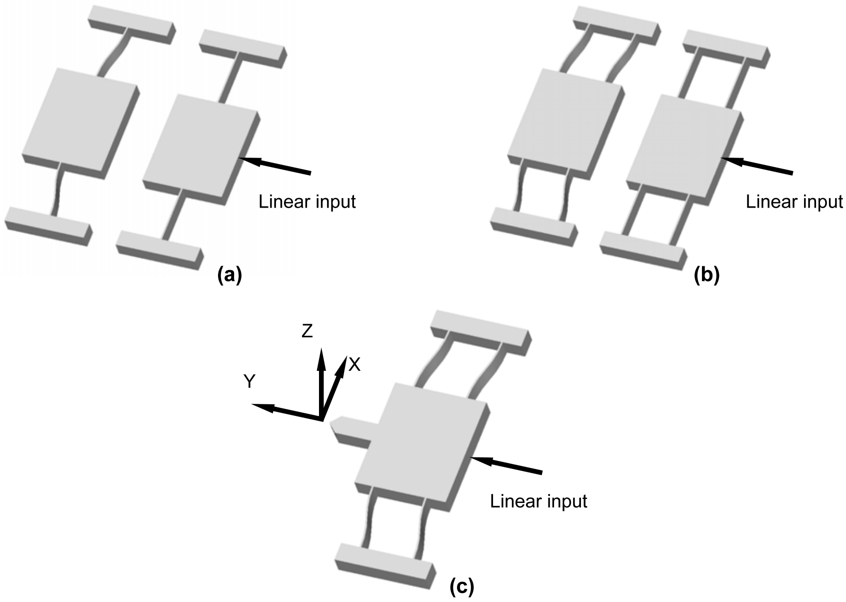

Figure 2. Flexure hinge illustration. (a) Beam-type flexure hinge; (b) parallel beam-type flexure hinge; (c) force-motion illustration.

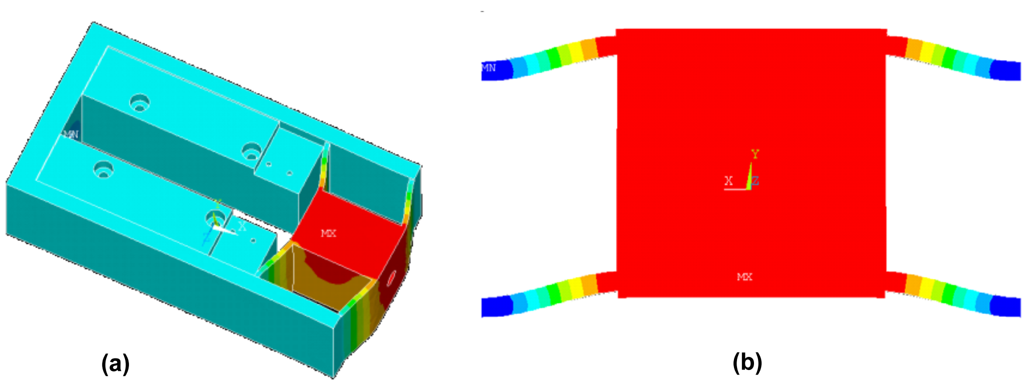

Figure 3. Static finite element analysis results. (a) Entire mechanism analysis; (b) flexure hinge analysis.

of piezoelectric actuation. It can be seen that, with the current flexure hinge design, the deformation is limited to the flexure hinge part only, while the housing part does not have significant deformation (figure 3). Therefore, we can focus our attention on the flexure hinge portion of the mechanism. The stiffness in the linear-motion direction (Y-axis) is calculated as 38 N μm^{-1}, which indicates that the motion stroke in this direction can be approximately preserved at $\sim$100 μm. The actual motion stroke depends on the preload and the output force of the piezoelectric stack actuator under operating conditions, which is subject to later experimental testing.

We further carry out modal analysis of the flexure hinge mechanism. Figure 4 shows the numerical analysis results of the first six mode shapes. The first mode is essentially in the linear-motion direction, with a natural frequency at 1131 Hz. The second mode represents the Y–Z twisting motion, with a natural frequency at 4054 Hz. The third mode is the X-axis linear motion, with a natural frequency at 4201 Hz. The other modes are in the twisting and torsional directions as well, with higher frequencies (7060, 7126, and 8147 Hz). These analysis results clearly indicate that the proposed flexure hinge mechanism, with the aforementioned design profile, can indeed: (1) avoid unwanted resonances in the low-frequency range; and (2) maintain an adequate motion stroke in the motion direction.

2.4. Prototype system

Based on the preceding design and analysis, a dual-stage actuation system prototype is fabricated, as shown in figure 5. Due to the monolithic structure character of the fixture, an electrical discharge machining (EDM) technique is employed in the fixture fabrication. A BEI Kimco Magnetics Division voice coil motor LA50 is used as the first stage actuator; it can provide a 50 mm motion stroke and a 300 N continuous stall force. As the motion guide of the first stage, a RAB6-3 air bearing (Nelson Air Corp) is used, which can provide over 2 000 000 lbs/inch stiffness within its motion range. As mentioned, a high-voltage piezoelectric stack actuator A050120 (Kinetic Ceramics Inc.) driven by a KC1000 amplifier (Kinetic Ceramics Inc.) is used as the second stage

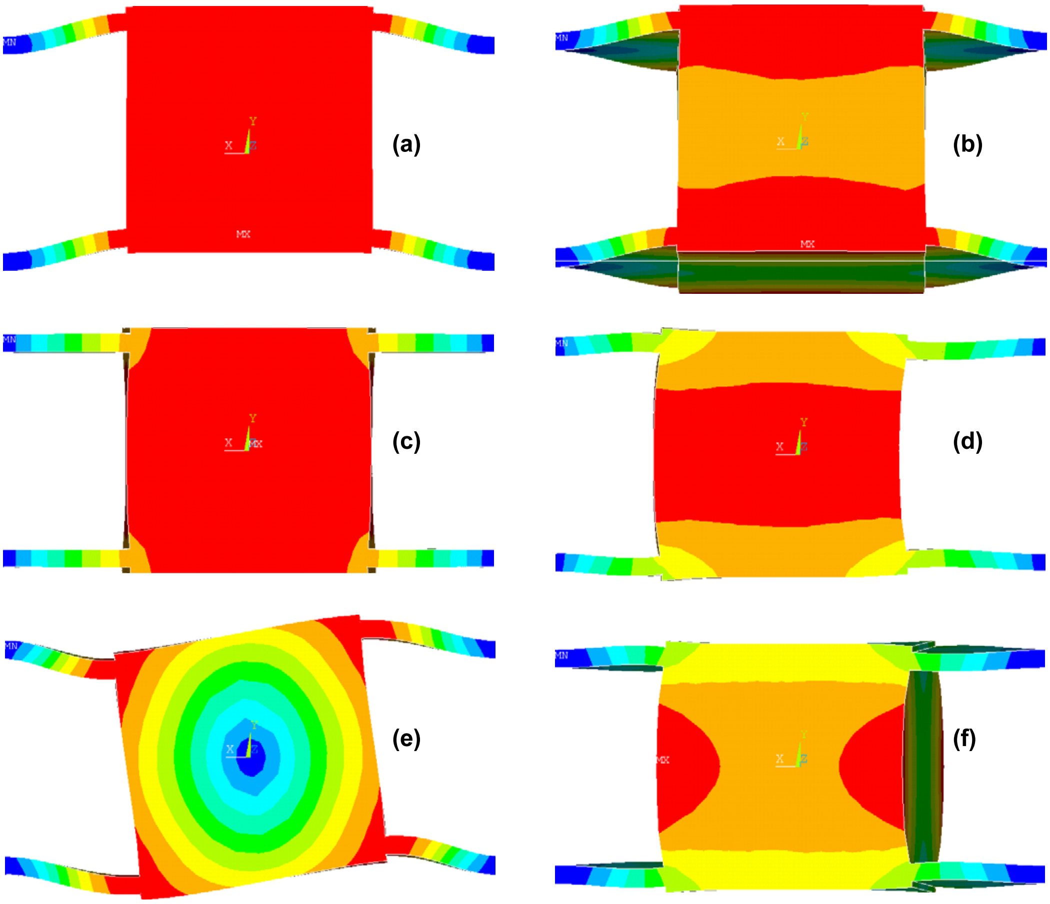

Figure 4. Dynamic finite element analysis results. (a)–(f) First six vibration modes.

actuator. An MS-80 linear encoder system (RSF Elektronic Inc.) is used as the first stage position sensing unit, which can provide a 10 nm resolution when a subdivision unit is connected between the sensor read head and the data acquisition board. For the second stage position measurement, a capacitive sensor (D510, PI Inc.) is used with a resolution and bandwidth of 2 nm and 10 kHz respectively over the 500 μm sensing range. For the preliminary control study, all these components are connected/commanded by a dSPACE DS1103 board (dSPACE Gmbh).

3. System identification and dynamic modeling

In order to carry out the control design, we need to identify the system characteristics and develop a mathematical model. The system characteristics, identified experimentally from the actual hardware, will also be used to verify whether the original design specifications are satisfied. The system identification is performed separately on two sub-systems: the VCM (with no input to the piezoelectric stack actuator) and the piezoelectric actuator housed in the flexure mechanism (with no motion of the VCM). A series of experimental modal analyses are carried out to obtain the frequency responses of these two sub-systems. We apply sine sweep actuations to the sub-systems respectively, and measure the corresponding outputs using the relevant sensors. The input–output relations in the frequency

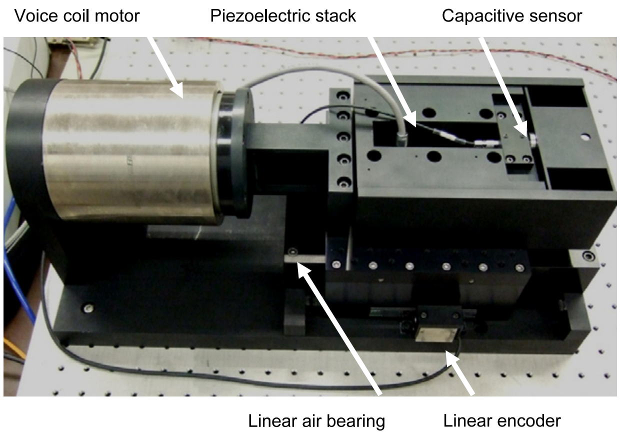

Figure 5. Dual-stage actuation system prototype.

domain are recorded by a signal analyzer, followed by the identification of the transfer functions using the curve fitting technique.

3.1. Experimental setup

The model identification experiments are carried out as follows. A dynamic signal analyzer (HP 35670A, Agilent Technologies) is used for frequency domain analysis. When

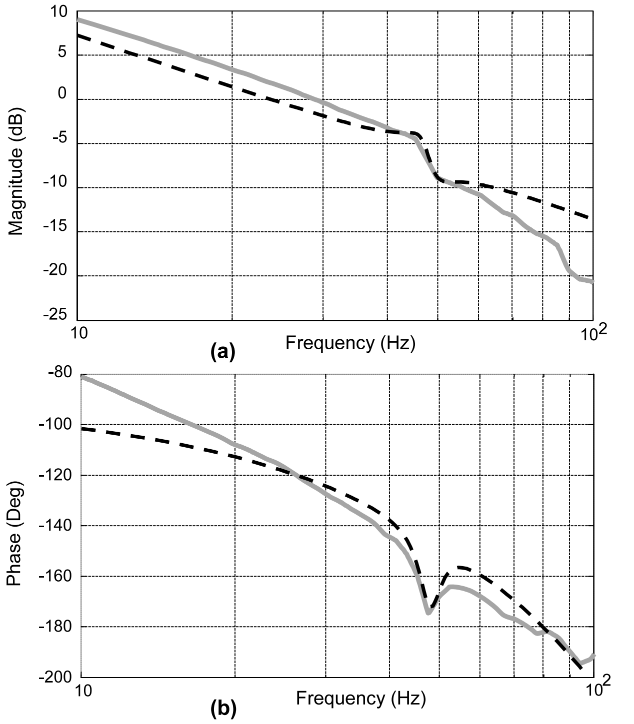

Figure 6. Frequency response of the VCM. (a) Magnitude; (b) phase. Solid: experimental result; dashed: identified model.

the fine actuation stage (the piezoelectric stack actuator housed in the flexure hinge mechanism) is tested and identified, only the piezoelectric actuator is actuated and the VCM is locked. The actuation voltage sent to the piezoelectric actuator amplifier is a swept sine signal. The displacement of the flexure hinge mechanism is then measured by the capacitive sensor, which is sent back to the dynamic signal analyzer. The signal analyzer produces the frequency response data, and these data are later used in the transfer function identification. For the coarse actuation stage identification, the VCM is actuated and the fine actuation stage is locked and fixed on the air bearing's sliding bar. The displacement (output) of the sliding bar is measured under the swept sine excitation (input), which can similarly yield the frequency domain information of the VCM stage. The experimental frequency responses of the two sub-systems are plotted in figures 6 and 7, respectively.

3.2. Frequency domain identification

While a variety of system identification algorithms exist, in this research we employ a relatively straightforward least squares curve fitting of the experimental Bode diagrams to identify the transfer functions, aiming at verifying the actuation system design and providing a basic model for preliminary control studies. The transfer functions of the two sub-systems that are identified should yield the minimization of the error function, i.e.,

$$\min\left(\sum_{n=1}^{N}\left\|\frac{Y_{\text{out}}(\mathrm{j}\omega_n)}{X_{\text{in}}(\mathrm{j}\omega_n)} - G(\omega_n)\right\|^2\right) \tag{1}$$

where $\frac{Y_{\text{out}}}{X_{\text{in}}}$ and G are, respectively, the transfer function identified and the frequency response measured. ω_n ($n = 1, \ldots, N$) are the frequency points in the frequency response measurement. The orders and the structures of $Y_{\text{out}}(\mathrm{j}\omega_n)$ and $X_{\text{in}}(\mathrm{j}\omega_n)$ should follow the physical characteristics of the system and also match the specific characteristics of the frequency responses.

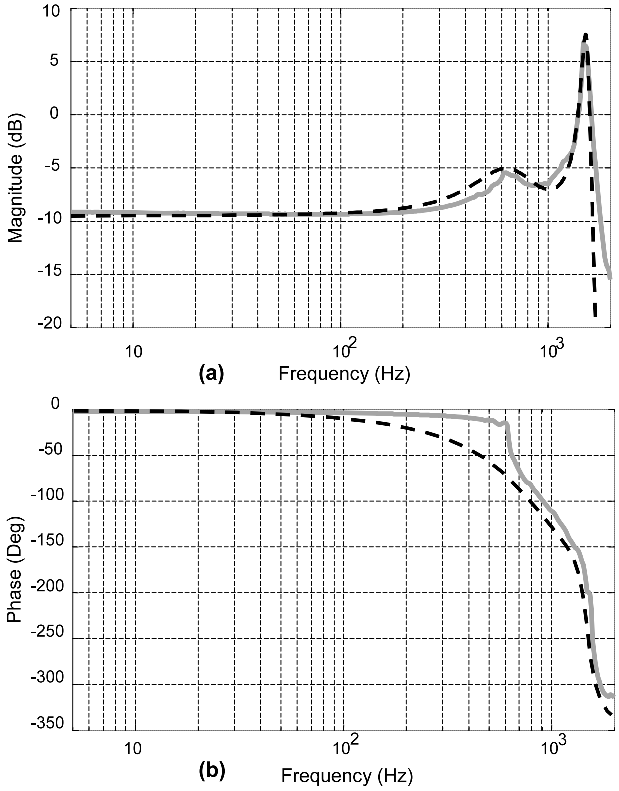

Figure 7. Frequency response of the piezoelectric actuator. (a) Magnitude; (b) phase. Solid: experimental result; dashed: identified model.

3.3. Identification results

The VCM response is measured in the frequency range 10–100 Hz with 100 measurement points. During the measurement, the piezoelectric stack actuator is locked. The measurement results are shown in figure 6, and are in qualitative agreement with results in the literature [21, 22]. Traditionally, the VCM system is considered as an electro-mechanical device with coupled electrical (voice coil circuit) and mechanical (mass and damping) sub-systems which form an internal closed-loop with a velocity–current feedback constant [21]. In addition, the VCM model should also include the power amplifier effect. These internal loops lead to multiple second-order effects in the apparent input–output relation of the VCM. In this paper, we use the formulation developed in [22], that explicitly relates the VCM system parameters to the transfer function, as the starting point of VCM identification. The actual mass, coil resistance and inductance of the VCM are incorporated in the transfer function relation. Curve fitting is then used to identify unknown parameters such as the damping and amplifier transfer function. The identified result is plotted in figure 6 and described by the following equation,

$$G_V(s) = K_V G_{V1} G_{V2} G_{V3} G_{V4} G_{V5}, \quad (2)$$

where

$$K_V = 25.03 \qquad G_{V1}(s) = \frac{1}{s}$$

$$G_{V2}(s) = \frac{\omega_{V2}^2}{s^2 + 2\varsigma_{V2} \cdot \omega_{V2} \cdot s + \omega_{V2}^2} = \frac{615\,411}{s^2 + 2272.2 \cdot s + 615\,411}$$

$$G_{V3}(s) = \frac{1}{\omega_{V3}^2} s^2 + 2\varsigma_{V3} \cdot \frac{1}{\omega_{V3}} \cdot s + 1 = 4.1 \times 10^{-4} s^2 + 3.4 \times 10^{-3} s + 1$$

$$G_{V4}(s) = \frac{\omega_{V4}^2}{s^2 + 2\varsigma_{V4} \cdot \omega_{V4} \cdot s + \omega_{V4}^2} = \frac{2218.4}{s^2 + 8.4s + 2218.4}$$

$$G_{V5}(s) = e^{-\tau s} = e^{-0.016s}.$$

In the above model, K_V is a constant gain, G_{V1} is the integral effect in the VCM, and G_{V2} comes from the electro-mechanical dynamic effect mentioned above. Since the system identification process is performed for the voice coil motor and the power amplifier that are combined together, the last three terms, including the time-delay effect, are attributed hypothetically to the unknown dynamic behavior of the power amplifier [21].

The frequency response of the piezoelectric stack actuator (housed in the flexure hinge mechanism) is measured in the frequency range 5–2000 Hz with 300 measurement points (figure 7). The response of the piezoelectric stack actuator can be modeled as a combination of one lightly damped mode at 826 Hz and one heavily damped mode at 1538 Hz. Thus, its transfer function can be identified as, after the least square based curve fitting,

$$G_P(s) = K_P G_{P1} G_{P2}, \quad (3)$$

where

$$K_P = 0.6299$$

$$G_{P1}(s) = \frac{\omega_{P1}^2}{s^2 + 2\varsigma_{P1} \cdot \omega_{P1} \cdot s + \omega_{P1}^2} = \frac{826.45^2}{s^2 + 2 \times 0.5901 \times 826.45 \cdot s + 826.45^2}$$

$$G_{P2}(s) = \frac{\omega_{P2}^2}{s^2 + 2\varsigma_{P2} \cdot \omega_{P2} \cdot s + \omega_{P2}^2} = \frac{1538.46^2}{s^2 + 2 \times 0.071\,29 \times 1538.46 \cdot s + 1538.46^2}.$$

The result of the identified model is plotted in figure 7 for comparison.

4. Preliminary control studies

Using the prototype hardware system and the system dynamic model identified, we carry out both open-loop control and

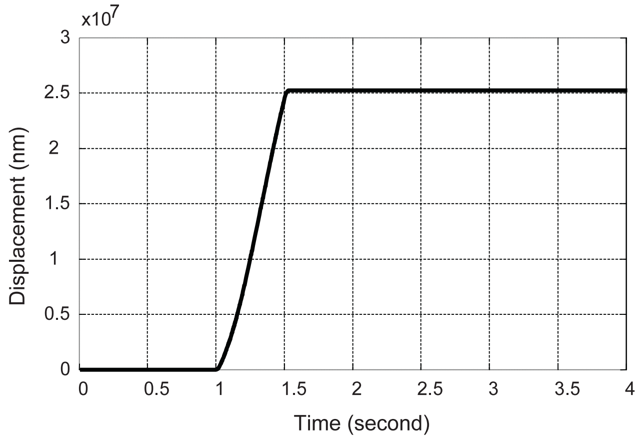

Figure 8. Open-loop experimental result of the VCM.

closed-loop control studies, which can further verify the system model and demonstrate the performance of the proposed design. The control experiments are performed using the dSPACE system. The hardware-in-the-loop results are compared with the numerical simulations under the same control parameters.

4.1. Open-loop experiments and analyses of individual actuation stages

In this section we analyze the open-loop system performance without feedback. Our purposes are: (1) to examine the characteristics of the two actuators under the dual-stage actuation system design; and (2) to verify the system model by comparing the open-loop experiment with numerical simulation. As will be shown in section 4.2, the preliminary closed-loop control presented in this paper is built upon a dual-loop strategy, i.e., each actuation stage is controlled using a classical PID controller. Here, in this section, we investigate the open-loop step responses of individual actuators which provide a performance baseline.

For the VCM sub-system, since there is an integrator in the model, the actual control voltage used in open-loop testing is a square wave with 0.5 V amplitude and 0.5 s duration. The experimental result and the simulation result are shown in figures 8 and 9, respectively. The VCM maintains a constant velocity during the square wave duration, and the final displacement is 25 mm. The experimental and simulation results agree with each other very well, which verifies the model validity from the time-domain response perspective.

We then investigate the piezoelectric actuator that is housed in the flexure mechanism. The displacement at the cutting tool location is measured. Therefore, the open-loop result includes the dynamic effect of the flexure mechanism. The open-loop results are plotted in figures 10 and 11, respectively. For a 5.5 μm step, the experimental result shows that the piezoelectric stack actuator has a settling time of 20 ms and 13% overshoot, while the simulation gives 25 ms settling time and 32% overshoot. It should be noted that the step response may actually include a higher-order dynamic effect that is not included in the model/simulation. Once again,

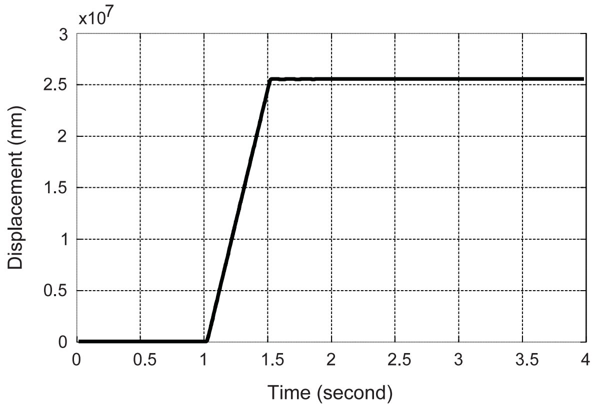

Figure 9. Open-loop simulation result of the VCM.

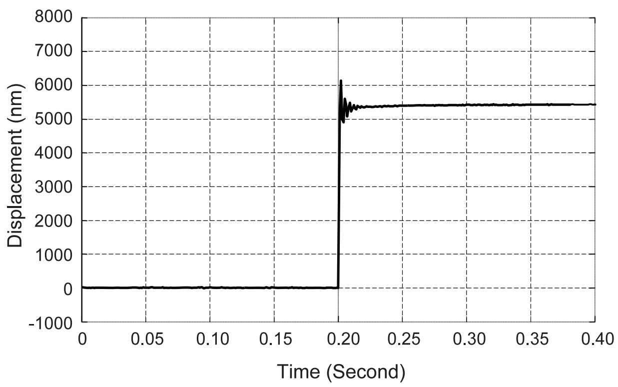

Figure 10. Open-loop experimental result of the piezoelectric actuator.

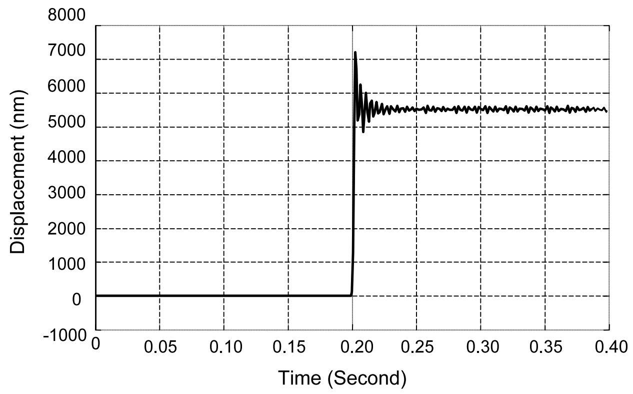

Figure 11. Open-loop simulation result of the piezoelectric actuator.

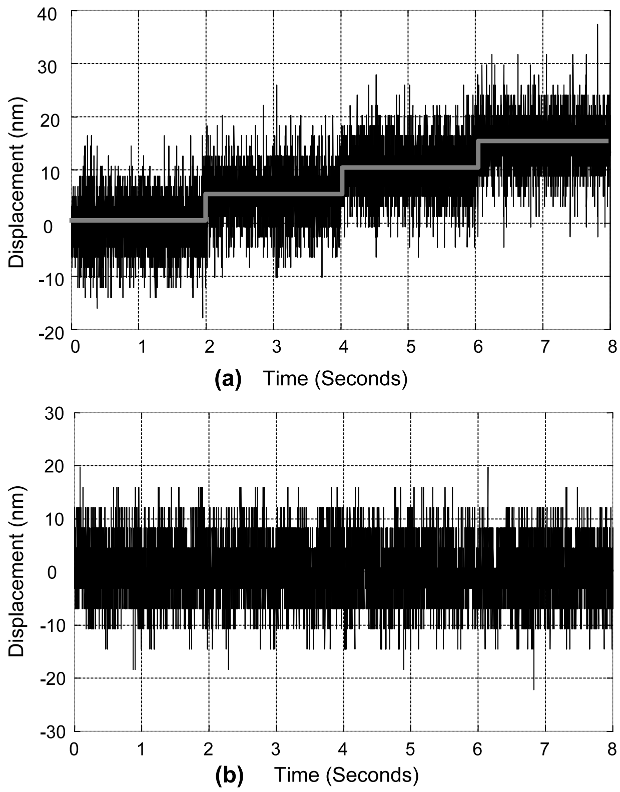

Figure 12. Open-loop motion resolution experiment of the piezoelectric actuator. (a) Sensor output under stepwise input; (b) sensor output without input.

this open-loop comparison shows reasonably good agreement between the model identified and the actual physical system.

One of the important characteristics of the piezoelectric actuator is its high resolution, which will be utilized to compensate the VCM. This high resolution directly influences the final performance of the dual-actuation system. Here we adopt the approach suggested in [23] to investigate the actuation resolution. We apply a series of stepwise inputs to the piezoelectric actuator. The corresponding output is shown in figure 12(a). For comparison purposes, the sensor output without the control voltage is shown in figure 12(b). It can be observed that the output displacement can follow the stepwise input, and the average resolution is better than 10 nm. This verifies that the fine stage piezoelectric actuator under the current housing design can indeed provide high actuation resolution.

Finally, the motion stroke of the piezoelectric actuator is also tested. The maximum stroke that the current system can yield is 88 μm. This is in agreement with the numerical prediction given in section 2.3. This stroke is adequate for the compensation of the VCM error.

4.2. Closed-loop control based on the two-loop strategy

The proposed dual-actuation system is a typical multiple-input–multiple-output (MIMO) control system. The primary goal of this paper is to illustrate the potential of the dual-stage concept as well as of the new hardware. Therefore, in this paper we employ a two-loop control strategy [24] based on ad hoc PID-type tuning to form a preliminary closed-loop system. This strategy is built directly upon the models of the sub-systems that have been identified. In this control scheme, the VCM coarse stage is utilized as the main control loop, and the piezoelectric actuator is commanded to compensate the error of the main loop (figure 13). The tracking reference signal is sent to the VCM control loop as input, and the error signal of the VCM position is sent to the piezoelectric actuator for high bandwidth and high resolution compensation. A saturation function is incorporated to take into account the limited motion

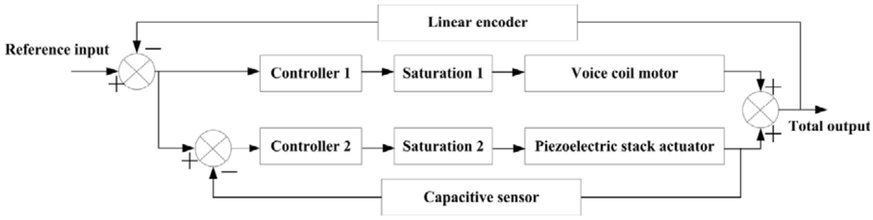

Figure 13. Closed-loop control structure illustration.

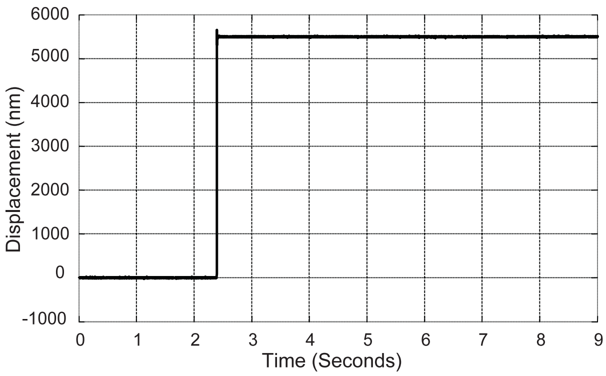

Figure 14. Closed-loop control experimental result of the piezoelectric actuator.

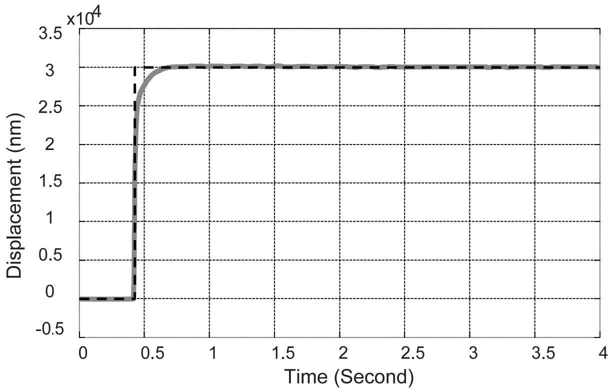

Figure 15. Closed-loop control experimental results: VCM only. Dashed: step reference; solid: VCM response.

range of the piezoelectric actuator. The overall output of the system comprises that of the piezoelectric actuator and that of the VCM. Owing to the feature of the proposed two-loop control structure, the control parameters are tuned separately for the individual stage/actuator. Obviously, such a tuning may not yield the optimal performance of the overall dual-stage system. Nevertheless, this control design can adequately utilize the system model identified, and lead to a direct comparison with respect to open-loop results of individual actuator stages to highlight the performance. Moreover, the results can form a baseline for future enhancement.

We first consider the coarse stage VCM actuator. The VCM is a position unstable actuator, because the input voltage leads to the velocity output. Here a traditional position–velocity (P–V) loop controller is used for the VCM position control [25]. In this controller design, the position loop compares a position command to a position feedback signal and calculates the position error. The position error is scaled by a constant to generate a velocity correction command that compares with the position differential values to generate the velocity error. Hence, a larger position error yields a larger velocity command. To achieve a smooth tracking, we incorporate a PI control for the VCM stage based on trial-and-error tuning, where $K_{P\mathrm{VCM}} = 110$ and $K_{I\mathrm{VCM}} = 15$. With this set of control parameters, the gain margin and phase margin are, respectively, 4.5 dB and 30°.

For the piezoelectric actuator, without loss of generality, here we utilize the Ziegler–Nichols method [26] for control parameter tuning. The tuning is carried out using the piezoelectric actuator sub-system model via simulation. We first disable the integrator term of the controller to get a pure P controller. We apply step input and keep increasing the P gain until an oscillation is achieved, which corresponds to a so-called ultimate gain. For this actuator, we have identified that $K_C = 0.8383$, and the corresponding oscillation period (ultimate period) $T_C = 0.75$ ms. The control parameters are then determined as $K_{P\mathrm{PZT}} = 0.45 \times K_C = 0.377$ and $K_{I\mathrm{PZT}} = 1.2 \times K_{P\mathrm{PZT}}/T_C = 600$. With this set of control parameters, the gain margin and phase margin are 6.24 dB and 60.1° respectively. The closed-loop performance of the piezoelectric actuator (with the flexure mechanism) is then experimentally examined and compared with its open-loop performance (shown in figure 14). Again we track the step input at 5.5 μm. Under the aforementioned control parameters, the settling time and overshoot are 10 ms and 2% respectively, which exhibits significant improvement over the open-loop results.

We finally investigate the closed-loop step tracking performance of the dual-stage actuation system. To examine the dual-actuation effect in particular, the reference is selected as a 30 μm step response. When the system is driven by the VCM alone (with the piezoelectric actuator locked), the settling time is 250 ms, as shown in figure 15. The error at steady state is 200 nm. When the system is driven by both actuators (dual-stage actuation), the piezoelectric actuator provides high bandwidth compensation to the VCM error, which is shown in figure 16. The error of the VCM only control and that of the dual-stage control are compared in figure 17. The settling time and the positioning error are significantly reduced to 50 ms and 20 nm, respectively. While this closed-loop control study is preliminary and there is plenty of room for performance enhancement, the improvement of

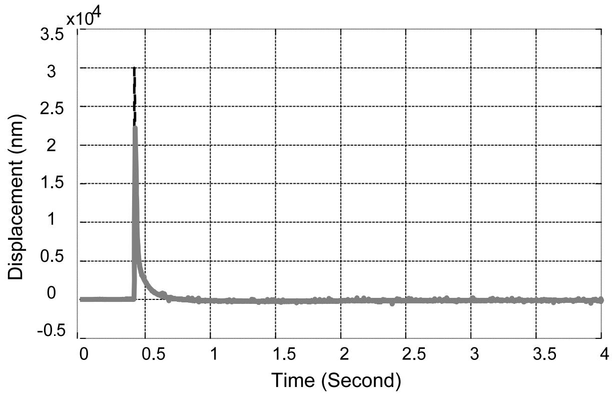

Figure 16. Closed-loop control experimental results: dual-actuation. Dashed: VCM error; solid: compensation of the piezoelectric actuator.

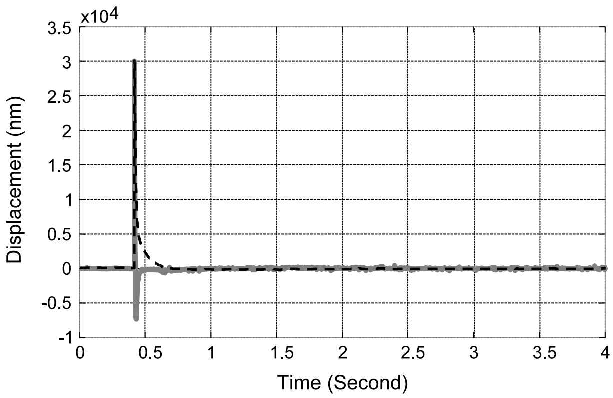

Figure 17. Closed-loop control experimental results: error comparison. Dashed: VCM only; solid: dual-stage control.

the dual-stage system over the VCM only system illustrated by the aforementioned investigations has demonstrated the effectiveness of the mechatronics synthesis and shown great potential for future research.

5. Concluding remarks

In this research, we have developed a dual-actuation system comprised of a VCM and a piezoelectric actuator for high precision and long stroke linear-motion control. Through comprehensive mechatronics synthesis, a parallel flexure hinge based mechanism is designed to provide a motion guide and preload to the piezoelectric stack actuator that is serially connected to the VCM. A series of numerical and experimental studies are carried out to facilitate system design, model identification, and control experiments. It is found that the fixture design can maintain an adequate resolution (better than 10 nm) and motion stroke (over 88 μm) of the piezoelectric stack actuator, which is adequate for future optical machining applications. The preliminary closed-loop experiment, using the dual-stage actuation system prototype, shows that the piezoelectric actuator can compensate the error of the VCM (improve the system precision from 200 to 20 nm), which could greatly enhance system performance. Future research will focus on a more sophisticated controller design that takes into account system nonlinearity and uncertainty and uses the actual tool-path as the tracking trajectory.

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